Inorganic Cumulative Exam
October 27, 2000

1. The ligands Cl^- and CN^- are often referred to as weak and strong, respectively. From a scientific perspective the terms "weak"and "strong'' are very poorly defined yet they continue to be used. To be more precise, the metal to ligand bond strength i.e., $\Delta H(M\text{-}L)$ would be preferred but it is rarely used. The scarcity is a practical issue and arises primarily from the difficulty of measuring a metal to ligand bond strength. Explain what difficulties are encountered when trying to measure metal to ligand bond strengths or describe any successful technique which can be used to determine the metal to ligand bond strength. Be very specific.

2. Despite the lack of specific direct measurements of metal to ligand bond strengths many experiments give indirect evidence for a relative ordering of bond strengths. Describe two such experiments and explain how the data can be used to say that one ligand is stronger than another.

3. Molecular orbital theory can be used to qualitatively describe the difference in ligand strengths of Cl^- and CN^-. Use MO diagrams to show how these two ligands interact with a metal in an octahedral ligand field. Show why CN^- produces a strong ligand field and Cl^- a weak ligand field.